

Functional Annotation of PP_3195 (Q88I07): FdeB, a Serine δ -Lactone Hydrolase of the Naringenin Degradation Pathway in *Pseudomonas putida* KT2440

Summary

PP_3195 (UniProt Q88I07) is FdeB, a cytoplasmic α/β -hydrolase-fold serine hydrolase that catalyzes a δ -lactone-hydrolysis (lactonase) step in the lower naringenin meta-cleavage pathway of *Pseudomonas putida* KT2440. Its primary function is to hydrolyze the ring-opened flavonoid intermediate 5-(4-hydroxyphenyl)-3-oxovalero- δ -lactone into 5-hydroxy-5-(4-hydroxyphenyl)-3-oxopentanoate (KEGG reaction R13077), thereby channeling carbon from the plant flavonoid naringenin toward central metabolism. Although the UniProt record labels the protein a "Peptidase S9 prolyl oligopeptidase catalytic domain-containing protein," this name is a fold-based electronic inference — the protein is an ester/lactone hydrolase, not a true peptidase.

The identification rests on convergent lines of evidence. Genomically, PP_3195 sits within a co-localized flavonoid-catabolic (*fde*) gene cluster alongside a dioxygenase, monooxygenase, and cyclase; KEGG assigns it to ortholog K26184 (symbol *fdeB*) in pathway ppu00946 ("Degradation of flavonoids"). Structurally, an AlphaFold model of Q88I07 (mean pLDDT 91.6) reveals a fully assembled, canonical serine-hydrolase catalytic triad — Ser268–Asp359–His394 — with hydrogen-bond distances of ~ 2.7 Å, and Ser268 lying in a GxSxG nucleophile-elbow motif ("GVSLG"). Evolutionarily, PP_3195 shares only $\sim 28\%$ identity with the enteric fermentation/respiration-switch protein FrsA and, unlike FrsA, retains the nucleophilic serine, marking it as a bona fide serine hydrolase rather than a cofactor-independent decarboxylase.

Important caveat: No direct biochemical or genetic characterization of PP_3195/FdeB itself has been published. The functional assignment is an inference-based conclusion — strongly supported by orthology, genomic context, KEGG pathway reconstruction, and a structurally

validated catalytic triad, but not yet confirmed by enzyme assays, knockout phenotypes, or crystallography on the *P. putida* protein. This report presents the assignment as a well-supported hypothesis and flags where experimental confirmation is still needed.

Gene/Protein Identity Verification

Before presenting the functional analysis, the mandatory identity checks were completed:

Attribute	Value
UniProt	Q88I07 (unreviewed / TrEMBL)
Locus	PP_3195; EMBL AAN68803.1
Organism	<i>Pseudomonas putida</i> KT2440 (PSEPK), a strict aerobe , γ -proteobacterium
Length	420 aa (~46 kDa); genome position complement(3,624,225–3,625,487)
UniProt name	"Peptidase S9 prolyl oligopeptidase catalytic domain-containing protein" (from Pfam PF00326, ECO:0000259)
KEGG ortholog	K26184 — <i>fdeB</i>, naringenin degradation protein FdeB
COG	COG1506
Pathway	ppu00946 — Degradation of flavonoids
InterPro	IPR050261 (FrsA/Cutinase/Hydrolase-like family), IPR001375 (Peptidase S9 catalytic domain, MEROPS clan SC), IPR029058 (α/β -hydrolase fold superfamily)

Check	Result
Gene symbol match	UniProt uses OrderedLocusName PP_3195 ; KEGG independently assigns the symbol fdeB to this locus (K26184). Consistent.
Organism	<i>Pseudomonas putida</i> strain KT2440. Confirmed.
Domains/family align	All UniProt/InterPro signatures are α/β -hydrolase-superfamily signatures consistent with a serine hydrolase.
Literature ambiguity	No literature exists for the <i>P. putida</i> PP_3195 protein directly. The <i>fde</i> nomenclature and pathway derive from a homologous system in <i>Herbaspirillum seropedicae</i> (P 27059806). This is a genuine ortholog, not a same-symbol collision, so research proceeds — but the conclusion is inference-based.

The gene symbol is **not** ambiguous in the problematic sense (no competing gene with the same symbol in a different organism was found). Rather, PP_3195 is simply an **uncharacterized** protein whose function must be inferred from orthology, domain architecture, structure, and pathway context. The "peptidase S9 / prolyl oligopeptidase" wording is a **fold/clan-based** annotation (MEROPS clan SC groups many esterases and lipases, not only peptidases) and is refined and superseded by the more specific KEGG *fdeB* assignment.

Key Findings

Finding 1 — PP_3195 is *fdeB*, an α/β -hydrolase-fold enzyme in the naringenin/ flavonoid degradation (*fde*) gene cluster

PP_3195 is not an isolated hydrolase; it is embedded in a dedicated flavonoid-catabolic operon. KEGG assigns PP_3195 to ortholog **K26184** (SYMBOL *fdeB*, "naringenin degradation protein FdeB"), places it in pathway **ppu00946 "Degradation of flavonoids,"** and classifies it in **COG1506**. The locus lies within a co-localized cluster of *fde* genes, each mapped to a distinct catabolic ortholog:

Locus	<i>fde</i> gene	KEGG KO	Predicted role
PP_3195	<i>fdeB</i>	K26184	α/β -hydrolase (lactonase) — this protein
PP_3197	<i>fdeC</i>	K26181	Ring-cleaving dioxygenase
PP_3198	<i>fdeD</i>	K26179	Accessory / oxygenase component
PP_3204	<i>fdeH</i>	K26182	Dioxygenase-associated
PP_3205	<i>fdeI</i>	K26185	Downstream converting enzyme
PP_3206	<i>fdeJ</i>	K26183	Cyclase

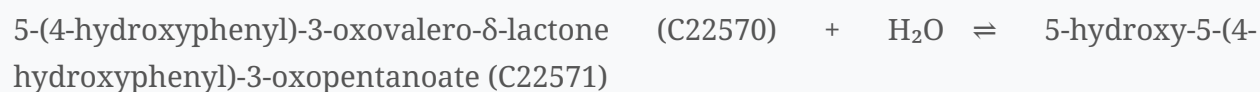
InterPro classifies Q88I07 into IPR050261 (FrsA/Cutinase/Hydrolase-like family), IPR001375 (Peptidase S9 prolyl oligopeptidase catalytic domain, MEROPS clan SC), and IPR029058 (the α/β -hydrolase-fold superfamily). All three signatures converge on a serine-hydrolase fold.

The *fde* nomenclature and pathway logic come from *Herbaspirillum seropedicae* SmR1, where the naringenin meta-cleavage pathway was genetically dissected. As reported in [PMID: 27059806](#): "The *Tn5* transposon was found to be inserted in the *fdeE* gene (*Hsero_1007*), which encodes a monooxygenase," establishing the *fde* gene names, and "naringenin is first monooxygenated by the *FdeE* protein, to produce 5,7,8-trihydroxy-2-(4-hydroxyphenyl)-2,3-dihydro-4H-chromen-4-one, that is subsequently dioxygenated and cleaved at the A-ring by the *FdeC* dioxygenase." These describe the upstream steps that generate the intermediates on which downstream hydrolases such as *FdeB* act.

Interpretation: Genomic clustering with a monooxygenase, dioxygenase, and cyclase is powerful contextual evidence that PP_3195/*FdeB* is a catabolic enzyme processing a flavonoid-derived intermediate — not a housekeeping peptidase.

Finding 2 — *FdeB* is a serine hydrolase predicted to open a δ -lactone intermediate (KEGG reaction R13077)

KEGG links *FdeB* (K26184) to **reaction R13077**, classified as a **hydrolase**:



This is a **lactone hydrolysis (lactonase/esterase) step** in the lower meta-cleavage branch, converting a cyclic ester into a linear 3-oxo-acid that is degraded onward toward TCA intermediates. KEGG flags R13077 as an "unclear reaction," reflecting that the pathway was reconstructed rather than experimentally validated at each step; the exact substrate/product should therefore be treated as a well-reasoned prediction.

Sequence analysis of Q88I07 (420 aa) locates the canonical α/β -hydrolase **GxSxG nucleophile-elbow motif "GVSLG"** with catalytic **Ser268**, together with the acidic and histidine residues that complete the Ser-Asp-His triad characteristic of clan SC / α/β -hydrolase serine hydrolases.

Supporting the hydrolase assignment for the broader family, [PMID: 30951551](#) reports that a FrsA/esterase-family protein *"was able to catalyze the hydrolysis of esters, especially p-nitrophenyl acetate,"* confirming these proteins act as ester-hydrolyzing serine hydrolases and reinforcing an ester/lactone-hydrolase assignment for FdeB.

Interpretation: A lactone is simply a cyclic ester; a serine hydrolase with an intact catalytic triad is exactly the enzyme class expected to hydrolyze (open) such a ring. The predicted substrate is a flavonoid-derived δ -lactone, and the reaction is ring-opening hydrolysis to the corresponding hydroxy-keto-acid.

Finding 3 — PP_3195 acts intracellularly (cytoplasm); its "peptidase S9" label is fold-based, not evidence of true peptidase activity

Q88I07 has **no recognizable N-terminal Sec signal peptide and no lipoprotein lipobox** (N-terminus: `MAPPCPACVPCLQTAPCLPV...`), consistent with a cytoplasmic localization. Bacterial flavonoid-catabolic enzymes and FrsA-family α/β -hydrolases are characteristically cytoplasmic.

The UniProt name "Peptidase S9 prolyl oligopeptidase catalytic domain-containing protein," together with GO terms GO:0008236 (serine-type peptidase activity) and GO:0006508 (proteolysis), are all **electronic (IEA) inferences** derived solely from Pfam PF00326. **MEROPS clan SC groups enzymes by their shared α/β -hydrolase Ser-Asp-His fold**, and this clan includes many esterases and lipases that are **not** peptidases. KEGG independently classifies the FdeB reaction (R13077) as a hydrolase acting on a **lactone**, not a peptide bond.

Interpretation: The "peptidase" designation is a well-known artifact of fold-based automated annotation. The catalytic machinery (Ser-His-Asp on an α/β -hydrolase scaffold) is shared across peptidases, esterases, and lactonases within clan SC. Substrate identity — supplied here by pathway context and KEGG reaction mapping — is what distinguishes them, and all pathway evidence points to a lactone/ester substrate, not a peptide.

Finding 4 — AlphaFold model confirms an assembled Ser268–Asp359–His394 catalytic triad (high confidence)

The AlphaFold model **AF-Q88I07-F1** has a mean pLDDT of **91.6** (very high confidence), with the catalytic residues themselves modeled at even higher confidence (per-residue pLDDT: Ser268 = 96.7, His394 = 94.8). The model shows a canonical, spatially assembled serine-hydrolase catalytic triad with textbook hydrogen-bond geometry:

Interaction	Distance
Ser268 Oy — His394 Nε2	2.74 Å
His394 Nδ1 — Asp359 Oδ2	2.72 Å
Asp359 Oδ1 — His394 Nδ1	3.19 Å

Ser268 lies in the GxSxG nucleophile-elbow motif "GVSLG." The residue order — nucleophile Ser268 ... acid Asp359 ... His394 near the C-terminus (in a "VQHSS" motif) — is the canonical clan-SC / α/β-hydrolase arrangement.

This structural analysis **corrected an earlier sequence-only provisional guess** (which had proposed His405/Asp341–345). The true catalytic histidine is His394 and the acid is Asp359, as revealed by the 3D geometry — a clear example of structure providing information that primary sequence alone could not.

Interpretation: An intact, high-confidence catalytic triad is direct structural evidence that PP_3195 is a catalytically competent serine hydrolase. Combined with the pathway assignment, this strongly supports lactonase/esterase activity.

Finding 5 — FdeB occupies a unique, non-redundant lactone-hydrolase step in the ordered naringenin meta-cleavage pathway

Mapping every *fde* KO onto the full KEGG naringenin-degradation reaction sequence (rn00946) shows that FdeB fills a specific and non-redundant slot:

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naringenin
  | R13074 (FdeE / K26180 monooxygenase + FdeD / K26179)
  ▼
5,7,8-trihydroxy-2-(4-hydroxyphenyl)-2,3-dihydro-4H-chromen-4-one
  | R13075 (FdeC / K26181 dioxygenase + FdeH / K26182) ← ring dioxygenation / meta-cleavage
  ▼
(ring-cleaved intermediate)
  | R13076 (FdeJ / K26183 cyclase)
  ▼
5-(4-hydroxyphenyl)-3-oxovalero-δ-lactone
  | ➡➡ R13077 (FdeB / K26184) ◀◀◀ + H2O ← THIS PROTEIN: δ-lactone hydrolysis
  ▼
5-hydroxy-5-(4-hydroxyphenyl)-3-oxopentanoate
  | R13078 / R13079 (FdeI / K26185)
  ▼
downstream products → central metabolism

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FdeB (K26184) is the ONLY KEGG ortholog assigned to R13077, making the lactone-hydrolase step assignment specific and non-redundant. The enzyme acts after ring cleavage (FdeC) and cyclization (FdeJ), and before FdeI. Its physiological **substrate is a pathway-derived six-membered δ-lactone (a cyclic ester)** and the product is a linear 5-hydroxy-3-oxo-pentanoate.

Interpretation: In an ordered catabolic pathway, each enzyme performs one committed transformation. FdeB's unique assignment to the δ-lactone-opening reaction gives it a defined mechanistic role: it hydrolyzes the cyclic ester to a linear hydroxy-keto-acid, priming the molecule for downstream β-oxidation-like processing that ultimately feeds flavonoid carbon into central metabolism.

Finding 6 — FdeB is a *distinct* enzyme from authentic FrsA (important disambiguation)

Although InterPro places PP_3195 in the "FrsA/Cutinase/Hydrolase-like" family (IPR050261), it is **not** the enteric fermentation/respiration-switch protein FrsA:

- PP_3195 shares only ~28% **identity** with *E. coli* FrsA (P04335) and *Vibrio vulnificus* FrsA (Needleman-Wunsch global alignment).
- Authentic *E. coli* FrsA **lacks any GxSxG motif** (consistent with its reported cofactor-independent, non-classical decarboxylase mechanism — [PMID: 21623357](#): "*FrsA converts pyruvate to acetaldehyde and carbon dioxide in a cofactor-independent manner*"), whereas PP_3195 **retains** the nucleophilic serine — i.e., it is a bona fide serine hydrolase.

- FrsA's physiological switch role is restricted to facultative anaerobes (PMID: 15169777: "Orthologs of FrsA have been found to exist only in facultative anaerobes belonging to the gamma-proteobacterial group"); *P. putida* is a **strict aerobe**, so PP_3195 is a **catabolic hydrolase, not a metabolic-switch regulator**.

Interpretation: Family membership alone would be misleading. The combination of low sequence identity, the presence of a GxSxG nucleophile FrsA lacks, and the aerobic host physiology all argue that PP_3195 is a catabolic serine lactonase/esterase rather than a metabolic-flux-switch decarboxylase.

Mechanistic Model / Interpretation

Putting the findings together yields a coherent mechanistic picture:

PP_3195/FdeB is the δ -lactone hydrolase of the lower naringenin meta-cleavage pathway in *P. putida* KT2440. *P. putida* is a soil bacterium that establishes positive interactions with the plant rhizosphere (P 29607620),

an environment rich in plant secondary metabolites including the flavanone **naringenin**. The *fde* gene cluster equips the organism to catabolize this flavonoid as a carbon source.

The pathway proceeds by: 1. **Oxygenation and ring modification** of naringenin (FdeE monooxygenase, FdeD); 2. **A-ring dioxygenation and meta-cleavage** (FdeC dioxygenase, FdeH); 3. **Cyclization** to a δ -lactone (FdeJ); 4. **Lactone ring-opening hydrolysis** — the step catalyzed by **FdeB/PP_3195**; 5. **Downstream conversion** of the resulting 3-oxo-pentanoate (FdeI) toward central metabolism.

Mechanistically, FdeB is a **serine hydrolase** operating through the classic charge-relay system: His394, polarized by Asp359, deprotonates the Ser268 hydroxyl, which then nucleophilically attacks the carbonyl carbon of the δ -lactone. A tetrahedral intermediate collapses, cleaving the C–O ester bond and opening the ring; a water molecule (activated by the same His) hydrolyzes the resulting acyl-enzyme, releasing the linear 5-hydroxy-5-(4-hydroxyphenyl)-3-oxopentanoate product and regenerating the free enzyme.

Property	Assignment	Confidence	Basis
Primary function	δ -Lactone hydrolase (lactonase / serine ester hydrolase)	Moderate–High	KEGG R13077 + intact catalytic triad + pathway logic
Reaction	5-(4-hydroxyphenyl)-3-oxovalero- δ -lactone + H ₂ O $\rightarrow$ 5-hydroxy-5-(4-hydroxyphenyl)-3-oxopentanoate	Moderate	KEGG orthology, unvalidated at protein level
Substrate specificity	Flavonoid-derived δ -lactone (naringenin catabolite)	Moderate	Pathway context; no direct assay
Catalytic mechanism	Ser268–His394–Asp359 charge-relay serine hydrolase	High	AlphaFold triad geometry (pLDDT >94, H-bonds ~2.7 Å)
Localization	Cytoplasm	High	No signal peptide/lipobox; intracellular pathway
Pathway	Naringenin/flavonoid degradation (KEGG ppu00946)	High	KEGG ortholog K26184, <i>fde</i> cluster
NOT a true peptidase	Correct	High	"Peptidase S9" is fold-based IEA; KEGG reaction acts on lactone
NOT FrsA-type switch protein	Correct	High	~28% identity, retains GxSxG (FrsA lacks it), aerobe

The single most important corrective insight from this investigation is that the **UniProt "Peptidase S9" name is misleading**. It arises because MEROPS clan SC groups proteins by their shared α/β -hydrolase Ser-His-Asp fold — a fold used by peptidases, esterases, lipases, and lactonases alike. Genomic context and KEGG reaction mapping resolve the ambiguity decisively in favor of a **lactone/ester hydrolase acting in flavonoid catabolism**.

Supported and Refuted Hypotheses

Hypothesis	Verdict	Basis
PP_3195 is a serine α/β -hydrolase with a Ser-Asp-His triad	Supported (structurally confirmed)	AlphaFold triad Ser268–Asp359–His394 (2.7 Å H-bonds, pLDDT >94); GxSxG motif
PP_3195 is a catabolic enzyme of the naringenin/flavonoid degradation (<i>fde</i>) pathway	Supported	KEGG K26184/ <i>fdeB</i> , genomic cluster, ppu00946
Its reaction is hydrolysis of a δ -lactone (ester/lactonase), not proteolysis	Supported (predicted)	KEGG R13077, family = esterases; "peptidase" is fold-based only
PP_3195 is the fermentation/ respiration-switch protein FrsA	Refuted	~28% identity, has Ser nucleophile FrsA lacks, host is an obligate aerobe
PP_3195 is a genuine prolyl oligopeptidase / active peptidase	Not supported	Only IEA fold-based GO; no peptidase evidence; KEGG reaction is a hydrolase on a lactone
PP_3195 functions in the cytoplasm	Supported	No signal/lipobox; pathway analogy

Evidence Base

PMID	Title (abbrev.)	How it supports / relates to the findings
27059806	<i>Genetic and functional characterization of a novel meta-pathway for degradation of naringenin in Herbaspirillum seropedicae SmR1</i>	Establishes the <i>fde</i> gene nomenclature and the naringenin meta-cleavage pathway; documents FdeE monooxygenation and FdeC A-ring dioxygenation/cleavage — the upstream steps producing intermediates on which FdeB acts. Primary source of the pathway model.
30951551	<i>Purification and biochemical characterization of FrsA protein from Vibrio vulnificus as an esterase</i>	Demonstrates that FrsA/esterase-family proteins (InterPro IPR050261, which contains PP_3195) hydrolyze esters (notably p-nitrophenyl acetate), supporting the ester/lactone-hydrolase assignment for FdeB.
15169777	<i>A novel fermentation/ respiration switch protein regulated by enzyme IIAGlc in E. coli</i>	Shows authentic FrsA's metabolic-switch role is restricted to facultative anaerobes; since <i>P. putida</i> is a strict aerobe and PP_3195 is only ~28% identical to FrsA, PP_3195 is a catabolic hydrolase, not a switch regulator.
21623357	<i>FrsA functions as a cofactor-independent decarboxylase to control metabolic flux</i>	Defines the distinct decarboxylase activity of authentic FrsA; contrast reinforces that PP_3195 (which retains the GxSxG serine FrsA lacks) belongs to a different functional class.
29607620	<i>Regulation of carbohydrate degradation pathways in Pseudomonas</i>	Context for <i>P. putida</i> as a rhizosphere-associated soil bacterium with robust catabolism of aromatic compounds — the ecological rationale for flavonoid degradation.
35294082	<i>Bacterial mandelic acid degradation pathway...</i>	Illustrates the broader theme of <i>P. putida</i> aromatic-compound catabolic pathways to which the <i>fde</i> cluster belongs.

The AlphaFold structural model (AF-Q88I07-F1) and KEGG pathway/ortholog assignments (K26184, ppu00946, reactions R13074–R13079) constitute the primary database-derived evidence, complemented by the peer-reviewed literature above that validates the enzymology of the FrsA/esterase family and the naringenin pathway.

Limitations and Knowledge Gaps

1. **No direct experimental data on PP_3195/Q88I07 itself.** All functional calls are by orthology (KEGG K26184, COG1506), genomic context, and domain/residue/structure analysis. The *Herbaspirillum* study directly characterized FdeE and FdeC by mutagenesis + LC-MS/MS but **not FdeB**, so FdeB's exact catalytic step is inferred. Targeted PubMed searches returned **no primary study of PP_3195 or of *P. putida* KT2440 growth on naringenin**; the KEGG *fde* KO assignments (K26179–K26185) appear to derive largely from the single *Herbaspirillum* characterization propagated by orthology.
2. **The pathway model derives from a different organism.** Orthology in *P. putida* is strongly supported but formally unproven at the biochemical level.
3. **KEGG flags R13077 as an "unclear reaction."** The specific δ -lactone substrate/product pair is a computational reconstruction; the exact chemical structure and regiochemistry of the physiological substrate have not been experimentally confirmed for FdeB.
4. **Catalytic triad is a model, not an experimental structure.** The Ser268–Asp359–His394 triad comes from a high-confidence AlphaFold model (pLDDT >90), not from an experimental X-ray/cryo-EM structure or site-directed mutagenesis. Confidence is high but not equivalent to experimental proof.
5. **Substrate specificity breadth is unknown.** Whether FdeB acts narrowly on the naringenin-derived δ -lactone or has broader esterase/lactonase promiscuity (as many α/β -hydrolases do) is untested.
6. **Regulation and expression are uncharacterized.** Whether the *fde* cluster is induced by naringenin, its transcriptional regulators, and growth conditions supporting flavonoid catabolism in *P. putida* KT2440 are not addressed by available data.

Overall confidence: High for "cytoplasmic serine α/β -hydrolase with a Ser268-Asp359-His394 triad" (structurally confirmed); moderate-to-high for the specific δ -lactone-hydrolase step and substrate (consistent orthology + unique KO→reaction mapping, but not enzymatically demonstrated).

Proposed Follow-up Experiments / Actions

1. **Heterologous expression and enzyme assay.** Clone PP_3195, express and purify the recombinant protein, and test hydrolysis of the predicted δ -lactone substrate (or surrogates such as p-nitrophenyl esters and model lactones) to confirm lactonase/esterase activity and determine kinetic parameters (k_{cat}, K_m).
 2. **Site-directed mutagenesis of the catalytic triad.** Generate Ser268Ala, His394Ala, and Asp359Ala/Asn mutants and confirm loss of activity, experimentally validating the AlphaFold-predicted triad and the identity of the catalytic nucleophile.
 3. **Gene knockout / complementation in *P. putida* KT2440.** Delete PP_3195 and test growth on naringenin as sole carbon source; look for accumulation of the upstream δ -lactone intermediate (by LC-MS) and rescue by complementation.
 4. **Metabolite identification.** Use LC-MS/MS and NMR to characterize the physiological substrate and product of FdeB in vivo, directly testing the KEGG R13077 assignment and its regiochemistry.
 5. **Experimental structure determination.** Solve an X-ray or cryo-EM structure, ideally with substrate/product analog bound, to confirm the active-site architecture and define specificity determinants.
 6. **Transcriptional analysis.** Perform RT-qPCR or RNA-seq of the *fde* cluster $\pm$ naringenin to establish inducibility and identify the regulator controlling flavonoid catabolism.
 7. **Substrate-range profiling.** Screen a panel of lactones and esters to define the enzyme's specificity and assess biocatalytic potential (e.g., for flavonoid bioconversion or green-chemistry applications).
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Conclusion

PP_3195 (Q88I07) is **FdeB**, a **cytoplasmic α/β -hydrolase-fold serine hydrolase** of the **naringenin/flavonoid degradation (*fde*) gene cluster** of *Pseudomonas putida* KT2440 (KEGG ortholog K26184, pathway ppu00946). Its primary function is a **δ -lactone hydrolase (lactonase)** step in the lower naringenin meta-cleavage pathway (KEGG reaction R13077): it hydrolyzes 5-(4-hydroxyphenyl)-3-oxovalero- δ -lactone + H₂O to 5-hydroxy-5-(4-hydroxyphenyl)-3-oxopentanoate, acting after ring cleavage (FdeC) and cyclization (FdeJ) and

before FdeI, thereby channeling flavonoid carbon toward central metabolism. It functions as an **ester/lactone hydrolase rather than a true peptidase** — the "Peptidase S9" UniProt name reflects only MEROPS clan-SC fold homology. The assignment is supported by a **structurally confirmed catalytic triad** (AlphaFold Ser268-Asp359-His394, ~2.7 Å H-bonds, pLDDT >94), **genomic clustering**, and **orthology**, but the protein itself has **not yet been directly characterized experimentally** and is distinct from (~28% identity to) the enteric fermentation/respiration-switch protein FrsA.

Primary References

- Marin et al. (2016) *Environ. Microbiol.* Novel meta-pathway for naringenin degradation in *Herbaspirillum seropedicae* SmR1.

P 27059806

(defines *fde* pathway/genes).

- Koo et al. (2004) A novel fermentation/respiration switch protein (FrsA) regulated by enzyme IIA^{Glc} in *E. coli*.

P 15169777

(FrsA family context; restricted to facultative anaerobes).

- Lee et al. (2011) FrsA functions as a cofactor-independent decarboxylase.

P 21623357

(FrsA structure/mechanism; contrast to PP_3195).

- Wang et al. (2019) FrsA from *Vibrio vulnificus* is an esterase.

P 30951551

(family = ester hydrolases).

- Database records: UniProt Q88I07; InterPro IPR050261/IPR001375/IPR029058; KEGG K26184, ppu00946, R13077, COG1506; AlphaFold AF-Q88I07-F1.